SAPC Framework: A Self-Adaptive Predictive Control Framework for Intelligent Air Pollution Management

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***Abstract* —** Air pollution has become a serious issue affecting both the environment and human health, particularly in urban and industrial regions[1]. Many current pollution monitoring systems focus only on measuring environmental conditions and reporting them once pollution levels exceed safe limits. These systems do not provide predictive capabilities or automatic control actions.

In this study, a new system called SL-PCG (Self-Learning Pollution Control Grid) is proposed [4]. It is an AI-based system designed to predict pollution levels and automatically control environmental conditions before they become harmful. The system integrates environmental data collection, time-series forecasting techniques, rule-based control actions, and a feedback-based learning mechanism. The model uses both simulated and real-time sensor data such as AQI, temperature, humidity, and traffic density to detect short-term pollution trends [2]. By analyzing these environmental parameters, the system can predict potential increases in pollution levels ahead of time. If the predicted values reach or exceed predefined safety limits, the system automatically activates pollution control methods such as ventilation or air filtration systems.

The experimental results demonstrate that the proposed system improves response time and reduces the severity of pollution spikes compared to conventional monitoring approaches [8]. Early intervention helps maintain a more balanced and stable environmental condition [5]. The system is also scalable and can be integrated with IoT devices, making it suitable for smart city infrastructure, industrial environments, and institutional areas like university campuses.

Keywords — Artificial Intelligence, Pollution Monitoring, IoT Systems, Time-Series Prediction, Smart Cities, Environmental Control, Intelligent Systems.

INTRODUCTION

Air pollution is a serious issue that affects many countries around the world. It causes harm to both the environment and human health. The rapid growth of cities, increasing industrial activities, heavy traffic, and uncontrolled burning of fuels have significantly contributed to poor air quality. As urban areas continue to expand, managing and monitoring pollution becomes more challenging. Many current monitoring systems only measure pollution levels and display the information, but they do not provide any prediction or automatic control to reduce pollution.

Traditional air quality monitoring systems mainly work in a reactive manner. They detect high Air Quality Index (AQI) levels only after pollution has already crossed the safe limit. Because of this, actions to control pollution are taken too late. This delay can increase the exposure of people to harmful pollutants and may lead to serious health problems such as asthma and other respiratory diseases. With the advancement of technologies like Artificial Intelligence (AI) and Internet of Things (IoT), it is now possible to improve the way environmental monitoring systems work. These technologies allow systems to collect real-time environmental data and analyze it using intelligent algorithms. Predictive analysis can help estimate future pollution levels and allow preventive measures to be taken earlier.

To address this issue, this study proposes SL-PCG (Self-Learning Pollution Control Grid). The system is designed to predict pollution patterns and automatically manage environmental conditions. It combines environmental sensors, predictive data analysis, automated control systems, and IoT communication. When the system predicts that pollution levels may increase beyond safe limits, it automatically activates mitigation systems such as ventilation or air filtration units. This helps reduce pollution exposure and maintain better air quality in cities and industrial areas.

SYSTEM BACKGROUND

Air quality monitoring systems are commonly used to measure the concentration of pollutants in the atmosphere. Most traditional systems use fixed monitoring stations that contain sensors capable of detecting gases and particles such as carbon dioxide (CO_2), nitrogen oxides (NOx), and particulate matter. These systems continuously collect environmental data, send alerts when pollution levels become unsafe, and store historical records for environmental analysis and regulatory purposes. While these systems are helpful for monitoring air quality, their main role is limited to observation. One of the main problems with conventional monitoring systems is that they do not have the ability to predict pollution levels in advance. These systems detect dangerous pollution levels only after the Air Quality Index (AQI) has already crossed safe limits. As a result, actions to control pollution are taken only after the environment has already been affected.

Another limitation is that most of these systems do not include automatic control mechanisms. When harmful pollution levels are detected, the response usually depends on manual actions rather than automated solutions. This delay can reduce the effectiveness of pollution control efforts. Traditional monitoring systems also lack adaptive learning capabilities. They do not update their analysis methods based on newly collected environmental data. In addition, these systems are rarely connected with devices such as ventilation systems, air purification units, or industrial emission controllers. Without this connection, the monitoring system cannot actively improve air quality.

The proposed SL-PCG (Self-Learning Pollution Control Grid) framework aims to solve these issues by introducing an intelligent environmental management system. The framework integrates predictive analysis, automated decision making, real-time sensing, and adaptive learning into a single architecture. By combining these features, the system can detect possible pollution risks early and automatically activate control mechanisms to prevent severe pollution conditions.

PURPOSE AND SCOPE OF THE STUDY

The purpose of the SL-PCG framework is to develop a smart pollution monitoring and control system that uses Artificial Intelligence to predict pollution levels and manage environmental conditions automatically. The system works as a closed-loop environmental management platform that constantly monitors environmental parameters, processes real-time data, and predicts possible pollution changes using data analysis techniques. When the predicted pollution levels approach or exceed safety limits, the system automatically activates appropriate control mechanisms to reduce pollution.

To perform this task, the system monitors several environmental factors. These include the Air Quality Index (AQI), pollutant gases such as carbon dioxide (CO_2), carbon monoxide (CO), and nitrogen oxides (NOx), along with other supporting parameters like temperature, humidity, and traffic density. These factors help the predictive model understand patterns and relationships that affect pollution levels. Based on this information and predefined safety limits, the system can automatically activate mitigation solutions such as ventilation systems, air purification units, exhaust fans, or alert notifications for relevant authorities.

The scope of this project includes designing and developing the entire SL-PCG framework. This involves creating the system architecture, developing the AI-based prediction model, implementing automated control logic, and building a dashboard interface for real-time monitoring and visualization. In addition, simulation experiments are conducted to analyze the performance of the system under different environmental conditions. The evaluation focuses on factors such as prediction accuracy, system response time, and reliability of operation.

CHALLENGES IN CONVENTIONAL POLLUTION MONITORING SYSTEMS

Conventional pollution monitoring systems are mainly designed to observe environmental conditions rather than actively control them. These systems detect pollutants only after their concentration exceeds safe limits. Because of this reactive nature, pollution control actions are taken only after harmful conditions have already occurred. This delay can increase the duration of exposure to polluted air and reduce the effectiveness of environmental protection efforts. A major limitation of these systems is the absence of predictive capabilities. Most monitoring platforms only measure pollutant concentrations and generate alerts when the values cross safety thresholds. They do not use data analysis or forecasting methods to predict future pollution levels. As a result, it becomes difficult for environmental authorities to take preventive steps before pollution levels become critical.

Another challenge is the lack of automated mitigation mechanisms. In many traditional systems, corrective actions are carried out manually after an alert is generated. This reliance on human intervention can slow down the response time and affect the efficiency of pollution management. Additionally, these systems generally do not include adaptive or self-learning features that allow them to improve their performance over time. Air pollution is influenced by several factors such as vehicle traffic, industrial emissions, and weather changes. However, many monitoring systems analyze pollution data as isolated readings instead of studying the overall environmental patterns. The limited integration between sensors and pollution control devices also prevents automatic responses. Due to these limitations, conventional monitoring systems often respond slowly, which may lead to higher health risks and less effective pollution management.

LIMITATIONS OF EXISTING APPROACHES

Although many environmental monitoring systems are currently available, they still have several important limitations. One of the main problems is the lack of predictive analysis. Most existing systems only monitor current environmental conditions and display pollution levels. They do not use forecasting techniques to estimate how pollution levels may change in the near future.

Another limitation is that monitoring systems and pollution control systems are usually developed separately. Some systems focus on collecting and displaying air quality data, while others are designed to improve air conditions through filtration or ventilation technologies. Since these components are not integrated into a single framework, the overall system cannot make intelligent decisions based on environmental data. Additionally, many traditional monitoring systems do not use advanced technologies such as Artificial Intelligence or machine learning. Because of this, they cannot continuously learn from new data or improve their prediction accuracy. Integrated systems that combine sensing, analysis, automated decision-making, and control actions are also uncommon. Due to this lack of integration and intelligence, many existing solutions are unable to provide effective and adaptive environmental control.

GAPS IN CONVENTIONAL ENVIRONMENTAL SYSTEMS

Conventional pollution monitoring systems still face several gaps that reduce their effectiveness in managing environmental conditions. One of the main gaps is the absence of a predictive-control architecture. Many monitoring systems collect real-time environmental data, but they do not integrate forecasting methods, automated mitigation actions, and intelligent decision-making within a single platform.

Another problem is the lack of adaptive learning mechanisms. Traditional systems usually rely on fixed models and do not adjust their performance as environmental conditions change. Without continuous learning, the system cannot improve its prediction accuracy or adapt to new environmental patterns. There is also limited coordination between sensing technologies and pollution control systems. While monitoring devices are capable of detecting pollution levels, they are often not connected to systems that can actively regulate environmental conditions. Because of this separation, proactive pollution control becomes difficult. These gaps indicate the need for a smarter environmental management solution. By integrating real-time monitoring, predictive analysis, and automated control within a unified data-driven system, environmental monitoring platforms can become more efficient and capable of managing pollution proactively.

IMPACT OF INEFFICIENT POLLUTION MANAGEMENT

Improper management of air pollution can have serious effects on public health, economic performance, and environmental sustainability. Continuous exposure to polluted air can lead to several health problems such as asthma, respiratory infections, and heart diseases. As air quality worsens, healthcare systems may experience increased demand, which raises treatment costs and creates additional challenges for society.

From an economic perspective, industries that fail to control emissions properly may face regulatory penalties or difficulties in meeting environmental standards. Poor air quality can also reduce employee productivity due to health-related absences and workplace disruptions. Additionally, inefficient pollution control practices may require more energy, resulting in higher operational costs and greater carbon emissions. Urban areas often face the greatest impact of these issues. When pollution levels increase and response measures are delayed, smog can accumulate and remain in the atmosphere for longer periods. This can create unsafe living conditions for people in these regions. These challenges highlight the need for proactive environmental monitoring systems that can predict pollution levels and implement control measures before the situation becomes critical.

SHIFT FROM REACTIVE MONITORING TO PREDICTIVE CONTROL

Traditional pollution monitoring systems mainly respond to environmental problems after they occur. However, the limitations of this reactive approach highlight the need for more intelligent environmental management systems. With recent advancements in automation, artificial intelligence, and data analytics, it has become possible to predict pollution levels and respond to potential risks at an earlier stage.

Predictive monitoring systems analyze both past environmental data and real-time sensor readings to understand pollution trends. By identifying unusual patterns or early warning signs, these systems can detect pollution increases before they reach dangerous levels. Once a potential risk is detected, the system can activate appropriate mitigation actions in advance. This approach helps reduce exposure to harmful pollutants and improves environmental safety. The proposed SL-PCG (Self-Learning Pollution Control Grid) framework is designed to support this transition from reactive monitoring to predictive environmental control. By combining predictive analysis, automated control systems, and adaptive learning, the framework provides a smarter and more sustainable method for managing air quality.

ROLE OF IoT TECHNOLOGY IN SL-PCG SMART POLLUTION CONTROL SYSTEM

IoT technology is an essential part of the SL-PCG framework because it enables real-time data collection, communication, and automated control of environmental conditions. In this system, various environmental sensors are used to measure parameters such as Air Quality Index (AQI), gas concentrations like carbon dioxide (CO_2), carbon monoxide (CO), nitrogen oxides (NOx), along with temperature, humidity, and traffic density. These sensors continuously gather environmental data and send it to a microcontroller-based edge processing unit. The edge device performs basic analysis and filtering of the sensor data before transmitting the processed information to a cloud platform.

The cloud system allows the storage and management of large volumes of environmental data and supports remote monitoring through digital dashboards. Users can view air quality data, predictive results, and system alerts using web or mobile interfaces. By combining sensing devices, cloud communication, data analysis, and automated control within an IoT architecture, the SL-PCG system converts traditional pollution monitoring into a smart and adaptive environmental management solution.

ADVANTAGES OF AN IoT-BASED SMART POLLUTION PREDICTION AND CONTROL SYSTEM

IoT-based pollution prediction and control systems provide several advantages compared with traditional pollution monitoring systems. One of the most important benefits is continuous real-time monitoring. Sensors placed in different locations collect environmental data regularly, which helps detect abnormal pollution conditions early. Another advantage is the use of predictive analytics supported by Artificial Intelligence. The system analyzes historical environmental data along with current sensor readings to understand pollution patterns and predict future changes. This allows pollution control mechanisms to be activated before pollution levels become dangerous.

The system also supports automated mitigation actions. Devices such as ventilation systems, air filtration units, exhaust controls, and alert systems can operate automatically when pollution levels rise. This automation reduces human involvement and improves response speed during critical situations. A centralized dashboard also makes it easier for users and environmental authorities to monitor pollution conditions in real time. By activating mitigation systems only when needed, the framework helps improve energy efficiency and reduce unnecessary power usage. At the same time, it supports environmental sustainability by lowering pollution exposure, reducing carbon emissions, and helping organizations meet environmental regulations. Overall, the system contributes to safer environments and more effective pollution management.

METHODOLOGY

The SL-PCG system is designed using a layered architecture that combines environmental sensing, intelligent data processing, cloud connectivity, and automated pollution control. This design allows the system to monitor environmental conditions continuously, analyze pollution patterns, and respond quickly when pollution risks are detected. By combining predictive analysis with automated control mechanisms, the framework supports proactive environmental management.

A. Environmental Data Acquisition Layer

The first layer of the system collects environmental information through various sensors. Gas sensors and environmental sensors are used to measure parameters such as carbon dioxide (CO_2), carbon monoxide (CO), nitrogen oxides (NOx), Air Quality Index (AQI), temperature, humidity, and traffic density.

These sensors detect environmental changes and convert them into digital signals. The sensor data is then sent to an edge computing device where it is prepared for further analysis. Continuous data collection helps identify pollution changes quickly and improves monitoring accuracy.

B. Data Preprocessing and Threshold Analysis Layer

The collected sensor data may contain noise or temporary fluctuations. These variations can occur due to traffic conditions, industrial emissions, or weather changes. To ensure reliable data, preprocessing techniques such as filtering, normalization, and calibration are applied.

After cleaning the data, it is compared with predefined safety thresholds. These thresholds help categorize pollution levels and determine whether the environmental condition is safe or hazardous. This analysis allows the system to detect critical pollution situations at an early stage.

C. AI Prediction and Adaptive Learning Layer

This layer introduces predictive capabilities using machine learning algorithms. The system studies both historical data and real-time sensor readings to understand pollution patterns. Time-series forecasting models are used to estimate future pollution levels. The system continuously updates its prediction model as new data becomes available. This adaptive learning process allows the system to improve prediction accuracy and adjust to changing environmental conditions.

D. Cloud Communication and User Interaction Layer

All processed environmental data and prediction results are sent to a cloud platform through IoT communication networks. The cloud system provides large-scale data storage and advanced analytics.

Users can access environmental data through a digital dashboard available on web or mobile platforms. The dashboard displays real-time pollution information, predicted trends, and system alerts, allowing users to monitor environmental conditions remotely.

E. Alerting, Decision Support, and Control Layer

The final layer of the system handles automated decision-making and control actions. By combining threshold analysis and AI predictions, the system determines when pollution control mechanisms should be activated. If pollution levels become unsafe, the system automatically activates devices such as ventilation systems, air filtration units, or exhaust control systems. Alert messages are also sent to users to inform them about environmental conditions. A manual control option is available for operators when direct intervention is required.

The overall SL-PCG framework follows a hybrid architecture that combines edge computing and cloud computing. Edge devices such as ESP32 microcontrollers handle real-time processing and immediate control actions, while the cloud platform manages long-term data storage and advanced analysis. This approach ensures both fast response and scalable system performance.

COMPARATIVE ANALYSIS

ASPECTS	CONVENTIONAL METHOD	PROPOSED SAPC FRAMEWORK
Prediction Capability	Does not support forecasting	Incorporates time-series prediction models
Response Speed	Delayed due to reactive approach	Faster due to predictive decision-making
Integration Level	Monitoring and control are separate	Fully integrated sensing and actuation system
Application Scope	Limited to observation purposes	Suitable for smart cities and industrial automation

BLOCK DIAGRAM

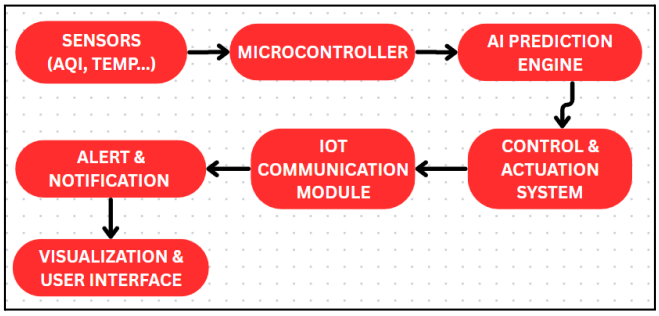


Fig.1: Block Diagram

IMPLEMENTATION

The SL-PCG system is implemented as a smart IoT-based environmental monitoring and control platform that enables real-time detection and mitigation of pollution. A network of environmental sensors is deployed across selected monitoring locations to measure key environmental parameters such as Air Quality Index (AQI), gas concentrations, temperature, humidity, and other indicators.

These sensors continuously transmit environmental data to an edge computing unit responsible for initial data processing. At the edge layer, operations such as noise filtering, data normalization, and threshold analysis are performed to prepare the data for further processing. The processed information is then used by AI-based predictive models that forecast short-term pollution levels. These models analyze both historical environmental data and real-time sensor inputs to identify patterns and predict potential pollution spikes. Depending on the system configuration, predictive computations can be performed either locally at the edge device or within the cloud environment.

The system uses a rule-based automation strategy to control pollution mitigation devices. When environmental conditions remain within safe limits, the mitigation devices remain inactive to reduce energy usage and extend equipment life. However, when pollution levels exceed predefined thresholds or predictive models indicate an upcoming pollution increase, the system automatically activates mitigation mechanisms such as ventilation units, air filtration systems, and exhaust control devices. Time-sensitive operations are executed at the edge layer to ensure immediate response to pollution events. The cloud infrastructure manages long-term data storage, system visualization dashboards, and advanced data analysis functions. To evaluate system performance, validation is performed using simulated environmental scenarios representing real-world conditions such as traffic congestion, industrial emissions, and changing weather patterns. Performance

metrics such as prediction accuracy, response time, system reliability, and operational robustness are used to assess the effectiveness of the system.

RESULTS

The experimental results indicate that the SL-PCG framework performs effectively in monitoring environmental conditions and predicting pollution levels. The implemented forecasting models are able to identify relationships between environmental parameters and pollutant concentrations. Based on these patterns, the system can predict short-term pollution changes and detect possible pollution spikes in advance.

Because of this predictive capability, mitigation mechanisms can be activated before pollution levels become dangerous. This proactive approach helps reduce the risk of prolonged exposure to high pollution levels. The use of edge-based automation ensures that pollution detection and response actions occur with minimal delay. This rapid response capability improves the overall efficiency of the system. At the same time, the cloud dashboard provides real-time access to environmental information such as AQI levels, gas concentration readings, prediction trends, and the operational status of control devices.

Long-term system testing under changing environmental conditions demonstrates that the architecture remains stable and reliable. The automated actuation mechanism also reduces unnecessary device operation, leading to improved energy efficiency. These findings show that the SL-PCG system can support efficient environmental monitoring while promoting sustainable pollution control practices.

CONCLUSION

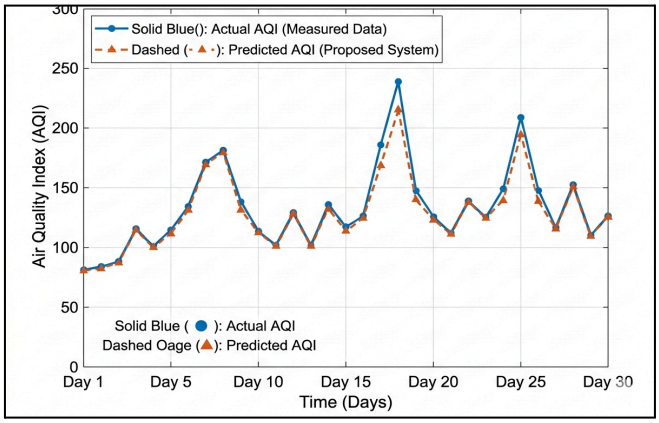
This study presents the SL-PCG framework as a smart and scalable solution for predictive pollution monitoring and control. By integrating IoT-based environmental sensing, artificial intelligence forecasting, automated decision-making, and mitigation mechanisms, the system enables efficient and proactive pollution management.

Unlike traditional monitoring systems that operate reactively, the SL-PCG framework predicts potential pollution events and activates control mechanisms before pollution levels reach hazardous limits. The hybrid edge–cloud architecture ensures quick local responses while allowing centralized data storage and advanced analytics through cloud services.

The proposed system demonstrates potential for deployment in various environments including industrial facilities, smart city infrastructures, academic campuses, and indoor environmental monitoring systems. Future development may focus on incorporating advanced deep learning techniques to improve prediction performance, integrating the system with smart infrastructure networks, enabling collaborative monitoring across multiple locations, and supporting data-driven environmental policy implementation to strengthen sustainable environmental management.

GRAPH ANALYSIS

Figure 2 shows how the predicted AQI values from the proposed SAPC framework compare with the actual recorded AQI values. The predicted curve follows the real AQI trend quite closely, which shows that the time-series forecasting model is working well. There are some small differences during sudden pollution spikes, but the system still manages to detect the rising pollution trend before it peaks. This early warning allows the control mechanisms to activate in advance, which helps reduce the severity of pollution peaks and keeps the environment more stable overall.



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